Gwenfred Tham Zhi Yee

AI & Computer Vision Engineer

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EDUCATION

Asia Pacific University of Technology & Innovation (APU), Kuala Lumpur **Nov 2023 – Nov 2026**

Bachelor of Computer Science (Hons) in Artificial Intelligence

- Expected Graduation: Nov 2026
- Relevant Coursework: Deep Learning, Computer Vision, Artificial Intelligence, Data Analysis with Python, Research Methods

EXPERIENCE

AI Data Labeling Intern, bits.ai **Jun 2025 – Oct 2025**

- Annotated traffic-light colour, shape and turn-direction attributes across 10,000+ image frames of computer vision training data, maintaining a 99% auditor pass rate.
- Applied documented labelling specifications and visibility rules to exclude unidentifiable signals, meeting a daily target of 10 video clips (supplied as image frames at 50 FPS).
- Collaborated with team leads and peers in continuous calibration sessions to resolve labelling ambiguities and align on annotation standards.

PROJECTS

DeepPCB-AOI: PCB Defect Inspection Platform **2026**

Personal Project | Python, ONNX Runtime, FastAPI, pytest | github.com/Tofui03/DeepPCB-AOI

- Built an industrial Automated Optical Inspection (AOI) platform for IPC-A-600 compliance triage on SMT/bare-board PCB lines, running template-aligned differential subtraction and morphological analysis via ONNX Runtime at sub-4 ms per board (260+ FPS).
- Detected 6 defect classes (open, short, mousebite, spur, spurious copper, pin-hole) against IPC-A-600 Class 2/3 tolerances, with deterministic safety guardrails so fatal defects (open/short) never pass and an automatic line-halt when batch defect recurrence reaches 3 or more.
- Skipped LLM/VLM calls for boards with no defect candidates (<15 ms turnaround, zero token cost); validated the system with 84 passing tests.

BJ-POS: Offline Point-of-Sale & Inventory System **2026**

Client Project (in-store rollout Nov 2026) | TypeScript, Vue 3, Pinia, Electron, SQLite, Vitest | github.com/Tofui03/BJ-POS

- Built to digitalize a small hardware store in Keningau, Sabah, replacing paper-based bookkeeping with an offline-first touchscreen till that gives real-time visibility of stock and sales.
- Made stock checking the core feature: each sale deducts stock in real time (unit, kg, ft or metre) without overselling, refunds and voids restore it, and an inventory view flags low- and out-of-stock items against an adjustable threshold.
- Covered the daily workflow: cash, card, QR, bank-transfer and contractor-credit (hutang) payments, thermal receipts, A4 delivery orders, and Close Day Excel reports backed up to Google Drive.
- Stored money as integer cents and quantities as milli-units to avoid rounding errors, and serialized database writes through one IPC queue so transactions never interleave; tested with Vitest.

Shipping Document Verification Platform (SDOC) **2026**

Hackathon Project | Python, FastAPI, Pydantic, React, pandas, pytest | github.com/Tofui03/sdoc-shipping-verification

- Developed and deployed on Render a FastAPI audit platform with a React web console that checks Shipping Instructions against draft Bills of Lading, using an LLM for email intent classification and OCR.
- Kept all 7 shipment field comparisons deterministic with MATCH / MISMATCH / NEEDS_REVIEW outcomes, retaining source evidence; reviewer corrections trigger revision checks and automatic re-comparison.
- Validated document extraction and data-transformation workflows with 307 passing pytest tests, using pandas v2 to structure test data.

SKILLS

Languages: Python, TypeScript, SQL, C++, Java, Bash

ML & Computer Vision: PyTorch, Torchvision, Ultralytics YOLOv8, OpenCV, scikit-learn, Hugging Face, ONNX Runtime, transfer learning, image classification, image annotation

Backend & Tools: FastAPI, Pydantic, React, Vue 3, Electron, SQLite, REST APIs, Gradio, Streamlit, pandas, NumPy, pytest, CUDA, Git, Docker, Linux/WSL

CERTIFICATIONS & LANGUAGES

Certification: Microsoft Generative AI for Beginners (course)

Languages: English, Malay, Mandarin